

5 Interesting Reference Designs For AUTOMOTIVE ELECTRONIC CONTROL UNITS

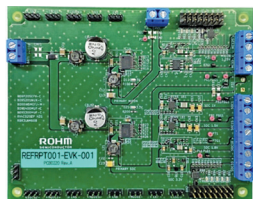
What if ECU design was already done? The right reference design can do more than save development time. Explore reference designs that help engineers build, test, and connect vehicle systems for power, control, lighting, and communication.

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Modern vehicles depend on numerous electronic control units (ECUs) to manage power, communication, safety, and other critical functions. Developing these systems from the ground up can be time-consuming and complex. The reference designs featured in this article offer ready-made platforms for building and testing ECU functions, including power management, engine control, lighting, communication, and security. You can use them to evaluate system architectures, connect vehicle subsystems, and verify their performance under real-world conditions. They also enable hardware and software validation before final products move into development.

Automotive ECU power management

This reference design helps automotive design engineers develop the power subsystem of infotainment and ADAS ECUs by providing a ready-made eight-circuit power tree with DC/DC converters, voltage monitoring, and functional safety support. You can use it to generate and manage the multiple voltage rails required by SoCs, microcontrollers, memory devices, CAN interfaces, and other ECU components, reducing the time



needed to design complex power architectures from scratch. The design also supports power sequencing, fault detection, and voltage supervision, helping improve system reliability and safety. By using this reference platform, you can evaluate automotive power-management strategies, accelerate ECU development, and create custom infotainment, ADAS, and vehicle control systems with a proven power-delivery foundation.

Applications. The reference design is tailored for infotainment devices such as automotive clusters, centre information displays, and ADAS ECUs.

Link. <https://www.electronicsforu.com/electronics-projects/reference-design-for-car-infotainment-and-ecu-systems>

OEM Brand. ROHM Semiconductor

Automotive electronic control unit design

This reference design helps design engineers rapidly develop and evaluate secure automotive ECU architectures for connected and safety-critical vehicles. Using the platform, you can prototype AUTOSAR-based applica-



tions, implement secure communication over CAN-FD networks, and validate message and node authentication to protect in-vehicle networks

from unauthorised access. The design also enables testing of cybersecurity functions such as secure boot, firmware authentication, trusted updates, and key management while supporting functional safety development aligned with ISO 26262 requirements. The included monitoring and intrusion-emulation capabilities allow you to analyse network traffic, test system resilience against cyberattacks, and accelerate the design of secure, reliable, and standards-compliant automotive electronic systems.

Applications. You can use it to develop and verify engine control, motor control, sensor interfacing, braking, airbag systems, climate control, infotainment, and other ECU applications.